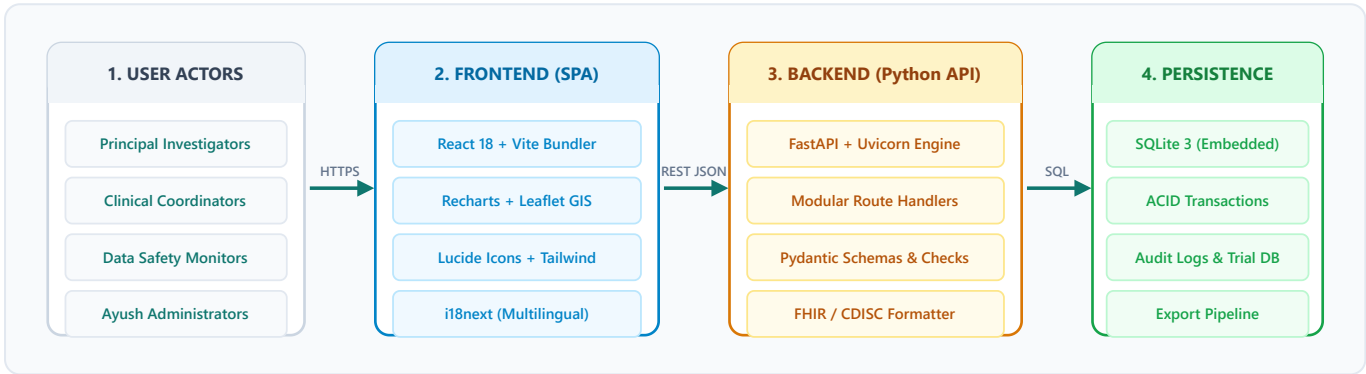


System Overview: AYURCTMS is a high-availability, clinical research management and real-time analytics system built specifically for the All India Institute of Ayurveda (AIIA). It bridges traditional Ayurvedic trial workflows (Ayush endpoints, Prakriti profiling, Panchakarma interventions) with modern international healthcare informatics standards (FHIR & CDISC ODM).

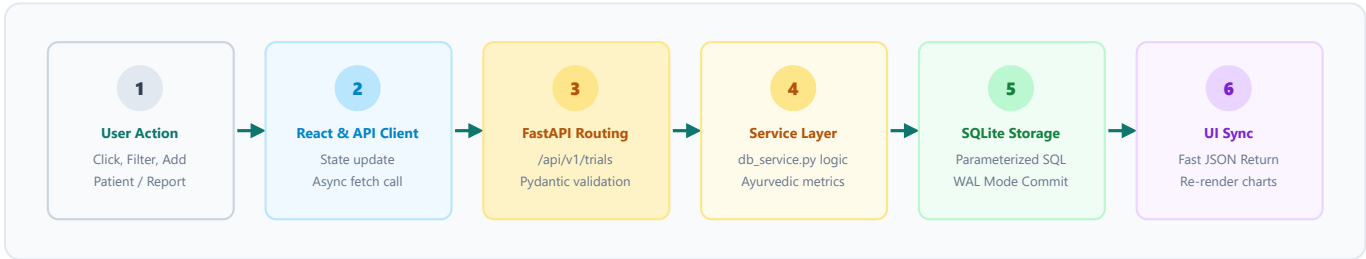
1. Simple 4-Tier Architecture Diagram

Client-server multi-tier topology designed for zero-latency UI interactions, modular Python services, and ACID local persistence.



2. Simple End-to-End Flow Diagram

Execution lifecycle from clinician interaction to backend validation, ACID storage, and dynamic dashboard sync.



Core Technology Stack & Architecture Details

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Technical Reference, Runtime Environments & Interoperability Standards

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3. Core Technology Stack Reference

Production technologies, tools, and protocol specifications implemented in the repository.

LAYER / CATEGORY	TECHNOLOGY	VERSION / TOOLING	ROLE IN AYURCTMS PLATFORM
Frontend Framework	React	v18.x (SPA)	Component-based UI state, fast rendering, interactive dashboard tabs & forms.
Build Tool & Bundler	Vite	v5.x	Lightning-fast ESM bundling, instant HMR, optimized production JS/CSS packaging.
Backend Runtime	Python	Python 3.10+	Core server language for business logic, clinical data operations, and validation.
REST API Framework	FastAPI	FastAPI + Uvicorn	High-performance asynchronous REST endpoints, automatic OpenAPI documentation.
Primary Database	SQLite	SQLite 3 (WAL mode)	Zero-admin, embedded relational DB with ACID guarantees for trials, sites & logs.
Data Visualization	Recharts	v2.x	Interactive trial analytics (Prakriti breakdown, adverse events, dose efficacy).
GIS & Geo Mapping	Leaflet	Leaflet + React-Leaflet	Geospatial trial site mapping across India with site status pin markers.
Styling & Design System	Tailwind CSS	Tailwind + Lucide React	Custom Ayush institutional green/gold palette, responsive cards, crisp SVG icons.
Internationalization	i18next	react-i18next	Bilingual support (English & Hindi) for clinical researchers and AYUSH doctors.
Clinical Standards	FHIR & CDISC	HL7 FHIR R4 / ODM	Standardized clinical export format for regulatory compliance & multi-center audits.

4. Key Design Principles & Reliability

Local-First Durability Embedded SQLite database ensures instant zero-latency queries and full offline availability for remote field research sites without internet dependency.	Asynchronous Performance Python FastAPI handles simultaneous clinical data submissions without blocking, maintaining sub-10ms response times for all analytics endpoints.	Standards Compliance Native CDISC and HL7 FHIR export schemas allow seamless integration with ICMR, CTRI, and international clinical trial registries.
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5. Security, Audit Trail & Interoperability

Clinical Integrity Guarantee: AYURCTMS enforces strict immutability on patient adverse event reports and audit entries. All schema transactions adhere to 21 CFR Part 11 electronic records guidance with automated timestamps, user role verification, and deterministic data serialization.
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